

--	--	--	--	--	--	--	--

***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2018***

**CS/EC/IT AS 15 -1102 B & CE/EE/ME/SE AS 15-1202 A ENGINEERING CHEMISTRY
(2015 Scheme)**

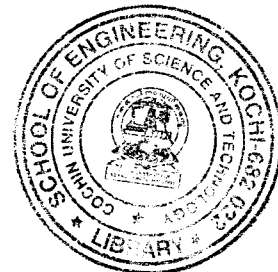
Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain why (i) Sodium chloride crystal is hard but brittle. (ii) Graphite is a conductor of electricity but diamond is a non-conductor.
 (b) Give the thermodynamics of adsorption.
 (c) TMS is used as internal reference in NMR. Why?
 (d) Derive Nernst equation for single electrode potential.
 (e) If 50% of a gas decomposed in second order way in 40 minutes, what would be the time taken for 75% of the gas to decompose?
 (f) Give the different statements of second law of thermodynamics.
 (g) Define the terms (i) Chemical potential (ii) fugacity.
 (h) State and explain Troutons rule. Give two exceptions to this rule with reason.
 (i) What are condensation polymerization and co-polymerization? Give one example for each.
 (j) Explain (i) aniline point (ii) viscosity index indicating their industrial importance.



PART B

(4 × 10 = 40)

- II. (a) What are point defects? Explain two stoichiometric and two non-stoichiometric defects in crystals with examples. (4)
 (b) The rotational spacing of (CN) in the microwave spectrum is 3.82 Cu^{-1} . Calculate the bond length. (C=12 N=14) (3)
 (c) How is BET adsorption isotherm used to determine the surface area of a solid catalyst? (3)

OR

- III. (a) Explain the formation of valence band and conduction band in sodium metal. How are solids classified on the basis of band theory? (4)
 (b) Explain the basic principle of NMR spectra. Indicate using a sketch the chemical shift and spin-spin splitting in the NMR spectrums of the molecule $\text{CH}_3\text{-CHO}$. (3)
 (c) Discuss the magnetic properties of solids. (3)
- IV. (a) Derive the integrated rate equation for the first order reaction. Give any three characteristics of first order reactions. (4)
 (b) What are the differences between chemical and electrochemical corrosions? (3)
 (c) In an electrochemical cell the cell reaction is $2\text{Ag}^+ + \text{Cu} \rightarrow 2\text{Ag} + \text{Cu}^{2+}$ (3)
 $E^\circ_{\text{cell}} = 0.46\text{V}$. Calculate the equilibrium constant of the cell reaction.

OR

(P.T.O.)

- V. (a) What are concentration cells? Give an example of electrotype concentration cell and derive an expression for its emf. How is electrolyte concentration cell used to determine the solubility and solubility product of AgCl? (4)
 (b) Describe H_2-O_2 fuel and give the cell reaction. State two advantages of fuel cells. (3)
 (c) Explain the determination of pH of a solution using quinhydrone electrode. (3)
- VI (a) Derive Kirchoff's equations. Mention one application of these equations. (4)
 (b) How is the function enthalpy derived? Define enthalpy of a reaction and enthalpy of neutralization. The enthalpy of neutralization of a strong acid and strong base is constant. Explain why. (3)
 (c) State Gibbs phase rule. Explain the phase diagram of water system using phase rule. (3)
- OR**
- VII. (a) Derive Gibbs-Helmholtz equation. Give the experimental verification of this equation. (4)
 (b) State reduced phase rule. Explain the phase diagram of Pb-Ag system applying reduced phase rule. (3)
 (c) Discuss thermodynamics of biological systems. (3)
- VIII. (a) Explain Cationic and coordination mechanism of addition polymerization. (4)
 (b) Give the structural difference between natural rubber and neoprene. Show the structure of vulcanized rubber. (3)
 (c) Write briefly on (i) Biodegradable Polymers (ii) Silicones. (3)
- OR**
- IX. (a) Explain any two moulding methods of plastic. (4)
 (b) Give any four properties of refractory materials. (3)
 (c) Describe the composition of Portland cement. Briefly explain the chemistry of setting and hardening of cement. (3)
